Coverage, Generalization, and Time: A Measured Study of Offline ARC-AGI-2

Two analyses — a coverage-vs-selection decomposition and a cross-family public-eval→leaderboard generalization gap — built on a clean single-GPU reproduction of the 2025 winner, with every claimed effect reported with its sample size and an explicit significance verdict.

ARC Prize 2026 — ARC-AGI-2 (offline track). Seventy-nine first-party leaderboard submissions: fifteen across five configurations, then a long same-code rerun series that measures leaderboard noise; see §3, §7 and §9. Linked submission: 33.06 (submission 57502566), the best of the 79 scored through Oct 26, 2026.

Abstract

We reproduce the 2025 ARC Prize winner (NVARC) as a dependency-light `transformers+peft` pipeline that runs on a single consumer GPU, and use it as an instrument to study what governs the offline ARC-AGI-2 score. We offer two analytical contributions. **First**, a *coverage-vs-selection decomposition*: by measuring the oracle (is the correct grid anywhere in the candidate pool?) against achieved pass@2, we show the dominant loss is **coverage** — the model never generates the right grid — not mis-ranking. Across five inference-time levers (selection tuning, a learned verifier, deeper decoding, a symbolic DSL solver, and 8B program induction with BARC) and four training-time levers (continued fine-tuning on locally-generated harder data, our own vocab-cut Qwen3-8B base trained on the same corpus, continued fine-tuning on 24k *frontier-LLM-generated* tasks loss-filtered to the slice the base does not know, and continued fine-tuning on 2.5B tokens of the

winner's own released corpus — §4.1), **none produces an effect distinguishable from zero at our sample sizes; the 8B base is directionally *worse*** (11.1% vs 22.8% per-task, paired n=30) **and the frontier-data run is exactly null** (44/133 vs 44/133 paired, 4 discordant each way — evidence that test-time training washes out base-model deltas); the largest positive point estimate (a deeper candidate pool, +2.4 pp paired) rests on a 3-vs-2 split of discordant outputs (exact McNemar p = 1.0) — and *inverted* on the real leaderboard (§7). This *is* the saturation result: test-time compute of the same kind does not add coverage on the hard eval distribution, and parameter scale without a deeper training corpus subtracts. The one exception is pooling *independent* runs, which lifts the oracle from 35.5% to 48.6% over three runs (n=138), while pass@2 rises only to 36.2%. Once the pool is that deep, selection becomes the bottleneck (§5). On the leaderboard, three draws of a public three-pass pooling kernel on the same model scored no better than single-pass reruns (§5). **Second**, a *generalization-gap calibration* across approach families, drawn from the competition forum: the LLM+test-time-training family loses ~3.5 pp absolute (~11% relative) from public eval to leaderboard, versus ~11 pp absolute (~51% relative) for a Tiny Recursive Model and a comparable collapse for a deep-recursive model — i.e. LLM+TTT is **uniquely leaderboard-robust**. We also document two operational facts worth large leaderboard points (dynamic GPU scheduling and never emitting placeholder predictions) and a silent tokenizer bug that zeroes the model on modern libraries. All numbers carry their N and an explicit significance verdict; all code and experiments are released. The study is now **leaderboard-confirmed** across 79 submissions (five distinct configurations plus repeated same-code draws that measure rerun variance — best 33.06 through Oct 26, 2026; the same-code series spans ~7 pp): our shallow-config submission scored **25.00** (semi-private set; a slightly *positive* eval→LB gap) and its byte-identical resubmit **24.17** — a first thin bound on same-code rerun noise (0.8 pp, n=2 runs) — while the deep-pool config, locally +2.4 pp, **regressed to 19.17**, far outside that bound. Two further submissions run the *same model checkpoint* through the community's public optimized engine and score **28.47 and 30.14**, and that engine's 71 later leaderboard draws centre on 29.3 (§7): **+4 to +5 pp bought purely by engine throughput** that converts the 12 h budget into per-task compute — the largest single effect in this study — while both fall ~4-5 pp short of that engine's advertised leaderboard score (§7). The two config points *straddle* the

family calibration, and their spread across two configurations of the same system exceeds the between-family gaps the calibration rests on — an instructive inversion analyzed in §6.1 and §7.

Contributions. (1) A faithful single-GPU reproduction of the winning recipe, plus a fix for a library-version tokenizer bug that silently scores 0. (2) The coverage-vs-selection decomposition and a measured saturation result over nine levers (five inference-time, four training-time), each with N and a significance verdict. (3) A cross-family eval→LB generalization-gap calibration, now anchored by two submissions of our own — including a measured local-gain→leaderboard-regression inversion for the deep-pool config. (4) A hardened offline submission engine (dynamic work-queue, no-placeholder policy). (5) A reproducible repository and an honest map of dead ends.

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1. Introduction

ARC-AGI-2 is engineered to be easy for humans and hard for machines, and was explicitly redesigned to resist brute-force program search; frontier models that score 20–50% on ARC-AGI-1 fell below 5% at its release. The Kaggle competition adds a hard operational constraint: submissions run **offline** on fixed hardware (L4×4, 96 GB, ≤12 h, no internet), excluding the frontier-API systems that top public ARC-AGI-2 leaderboards.

In 2025 one approach dominated the offline track: a small open LLM, fine-tuned on synthetic ARC data, **test-time-trained** per task, decoded into a candidate pool, and reduced to two answers by augmentation voting (NVARC, 24% private). The 2026 offline leaderboard is largely teams iterating on that recipe.

We do not propose a new architecture. We reproduce the winning recipe on a single consumer GPU and then **measure**, with sample sizes and confidence intervals, the levers practitioners reach for — reporting negatives as carefully as positives. The framing question is operational: *given a fixed strong base model and an offline budget, what actually changes the score?* Our answer is that it is governed far more by **coverage** (does the model generate the right grid at all),

generalization (does local accuracy survive the leaderboard), and **time** (does the run finish without placeholders) than by the decoding/selection tricks that attract most effort.

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2. Background and setup

Task & metric. Each task gives a few input→output grid demonstrations and one or more test inputs (grids ≤30×30, 10 colors). Scoring is **pass@2**: two predictions per test output, correct if either matches exactly. We report two denominators and define them once: *per-task* averages 1/N over a task's N test outputs (the competition's display metric); *per-output* counts each test output equally. The two differ by ~7 pp on our slice (≈1.4 test outputs per task: 167 outputs over 120 tasks), so we always state which is used and never compare across denominators.

The NVARC recipe (reproduced). Base: a Qwen3-4B fine-tuned with a 16-token grid vocabulary, publicly released by the 2025 winner. Per task: (a) **test-time-train** a per-task LoRA on the task's demonstrations under random augmentations (D4 geometry × color permutation); (b) decode a candidate pool with a batched **depth-first search** keeping token paths above a per-token probability threshold; (c) select two answers by **augmentation-NLL voting**. We re-implement this in stock `transformers+peft` (SDPA attention, gradient checkpointing) so it runs on one 32 GB GPU and on Kaggle L4s without `unsloth`/Flash-Attention/`bitsandbytes`; the decoder and scorer are ported faithfully from the winner's open notebook. Hyperparameters in Table 3.

A silent, fatal compatibility bug. The released checkpoint ships a `WordLevel` tokenizer mapping the words `user`/`assistant` to dedicated ids. Under `transformers ≥ 5.0`, `.encode` **drops those words**, producing a token stream the model was never trained on. The mechanism is unambiguous — `.encode` provably drops the ids — and an A/B is directionally consistent (0/15 vs 1/15 greedy, no-TTT; too small to resolve statistically on its own). We restore correctness with an exact custom encoder. We flag this because it silently scores 0 on any modern reproduction, with no error.

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3. Reproduction (local)

On the public ARC-AGI-2 evaluation set our port scores **22.8% per task / 29.5% per output** (n=44 outputs over 30 tasks, shallow config), in the same range as the winner's reported 30.5% pass@2 / 27% public-LB / 24% private-LB. Per-task latency is ~130 s (shallow) to ~261 s (deep) on one RTX 5090. **Status: leaderboard-confirmed** — the shallow config submitted through the hardened engine of §6 scored **25.00** on the leaderboard (semi-private set), a slightly *positive* gap vs our 22.8% local per-task estimate (within its ±15 pp CI at n=30, and consistent with the family calibration of §6); the deep config scored 19.17 (§7). Grand-Prize eligibility is satisfied (§9).

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4. The lever map — measured, with N and 95% CIs

All numbers are held-out pass@2 (per output) on public-eval candidate caches. "Oracle" = correct grid present anywhere in the pool (coverage ceiling). CIs are Wilson 95%. At our sample sizes the 95% CI half-width on a single proportion is **≈ ±10 pp (n=78)** to **≈ ±14 pp (n=42)** — so *pp-level differences below ~10 pp are not statistically resolved*. We report them anyway, with that caveat, because the **pattern of uniformly-near-zero effects is itself the result**.

Table 1 — inference-time levers.

Lever	Effect (point estimate)	N	Significant?	Verdict
Deeper pool (aug16 + DFS prob 0.1 + 3 views, *combined*)	**+2.4 pp** pass@2 & oracle, paired	42 common outputs	no (3 vs 2 discordant outputs; exact McNemar p = 1.0)	largest point estimate; not resolved
Free selection sweep (count-wt ×	+0 to +1 pp (≤1 task)	78	no	NVARC scorer already

Lever	Effect (point estimate)	N	Significant?	Verdict
{sum,mean,geom} aug-NLL)				near-optimal
Learned verifier (GBM): group-CV at 29 pos → train-on-1,000-training-tasks transfer at 1,037 pos	**−1.3 pp** → **+1.3 pp** (25→26 of 78)	78 test outputs	no (±1 output)	36× data flips the sign but adds ≤1 output; top feature is the aug-NLL rank kgmon already uses
Symbolic DSL (~15 rules, depth-2, train-consistent)	**0** / **167** outputs	167	yes (true 0)	ARC-AGI-2 defeats brute-force DSL
Program induction, three generators: BARC-8B (48–64 prog/task); Qwen2.5-Coder-7B-Instruct + repair round (8+6 prog/task); symbolic DSL (row below)	BARC: training **23%** (7/30 tasks), **eval 0/166** net-new; Coder-7B: **0 verified** / **120 eval** (best 1/5 train pairs, once)	166 / 120	yes (true 0 on eval, twice)	strong in-distribution (BARC); eval defeats execution-verified induction from every accessible generator

Two single-knob ablations: aug_n 8→16 *alone*, and “more views + lower threshold” *alone*, each measured **+0.0 pp oracle** (n=20–24). Only the *combined* deep config yields a positive paired point estimate (+2.4 pp,

n=42) — itself a 3-vs-2 split of five discordant outputs (exact McNemar p = 1.0). **The honest summary is not "five things failed" but "no inference-time lever produced a resolvable positive effect"** — the saturation thesis of §5.

4.1 Training-time levers: harder data, and a bigger base

If inference-time compute is saturated, the remaining local levers act on the *weights*. We measured both, paired on the same first-30 eval tasks and the same TTFT→decode→vote configuration as the baseline (per-task pass@2 22.8%).

Table 1b — training-time levers.

Lever	Effect (point estimate)	N	Significant?	Verdict
Continued-FT of the 4B on locally-generated *harder* data (compositional generator; 2k tasks, ~2× harder by loss: 0.10 vs 0.05)	**±0.0 pp** (identical 22.8%)	30 tasks, paired	no	locally-generated difficulty ≠ frontier-grade difficulty
8B base (vocab-cut Qwen3-8B, LoRA r=256 incl. embeddings, 1 epoch / 526M packed tokens of the same corpus, ~74 GPU-h)	**−11.7 pp** (11.1% vs 22.8%)	30 tasks, paired	no (within CI) but directionally negative	scale does not compensate for corpus depth
Continued-FT on *frontier-generated OOD*	**±0.0 pp** (44/133 = 33.1%		exactly null	even frontier-quality OOD

Lever	Effect (point estimate)	N	Significant?	Verdict
data: 24k GPT-4o-generated BARC-HEAVY tasks, loss-filtered to the slice the base has not absorbed (>0.10; 25% of 95k), + retention mix	both; discordant 4 vs 4)	133 outputs, paired		data does not move eval; TTT washes out base deltas
Continued-FT on the *winner's own released corpus* (2.5B tokens of the public 3.2M-task set, 19,637 steps, clean convergence)	**−4.6 pp** (29.5% vs 34.1%; discordant 4 vs 10)	129 outputs, paired	directionally negative	the released checkpoint sits at a selected optimum; more optimization on its own corpus *subtracts*

The 8B result deserves its cautionary tale. A first run at peak lr 2e-4 (the recipe's 1e-4, doubled) **diverged at step ~200** — loss 0.3 → 3.3, *above* the uniform-over-16-tokens entropy ln 16 ≈ 2.77 — despite gradient clipping at 1.0, and 46 further hours of cosine decay "recovered" it only into a degenerate basin (plateau 0.46; greedy decoding emits two cells then EOS; **empty candidate pools on 30/30 eval tasks**). A merge-vs-unmerged probe exonerated the checkpoint plumbing (teacher-forced loss 0.708 merged vs 0.711 unmerged) and isolated the fault to the weights themselves. The clean rerun at peak lr 5e-5 converged normally (final train-mix loss ~0.065) and produced a *functional* model — well-formed grids, healthy 133 s/task decode work, 4/30 tasks with at least one correct output (3.33/30 = 11.1% per-task with partial credit on multi-output tasks) — that is simply **weaker than**

the winner's 4B: teacher-forced loss on ARC-train examples 0.082 vs the 4B's 0.017. Our one-epoch pass over NVARC-synthetic + ARC-train + locally-generated data cannot match the corpus scale and training depth behind the released 4B checkpoint, and 2× the parameters do not close that gap; they appear to widen it at equal data.

The third row removes the last excuse. Where the harder-data row used *procedurally* generated tasks (which probed barely above the base's competence), this experiment used **frontier-LLM-generated data**: we streamed 95k GPT-4o-generated BARC-HEAVY tasks through the base and kept exactly the 24k the model measurably does not know (teacher-forced loss > 0.10; the kept slice averages ~3× the loss of NVARC's own synthetic corpus). Training was clean (loss 0.13 → 0.085 over 783 steps with a retention mix), and a 30-task A/B even flashed +1.6 pp — which the paired extension to **133 outputs erased exactly: 44/133 vs 44/133, four discordant outputs each way**. The FT model solves *slightly different* outputs, not *more* outputs. Our working hypothesis, consistent with §6's robustness mechanism: **per-task test-time training washes out base-model deltas** — whatever the base absorbs at training time, TTT re-adapts both models onto the same per-task solutions, so training-time interventions at locally-feasible scale shuffle the solved set without growing it. All three rows point the same way: **the binding constraint is not parameter count, not local data generation, and not even frontier-generated data at 10^4-task scale — new eval coverage evidently requires the winner-scale combination of corpus and training depth**, consistent with §5's conclusion that coverage requires a stronger base *trained on stronger data*.

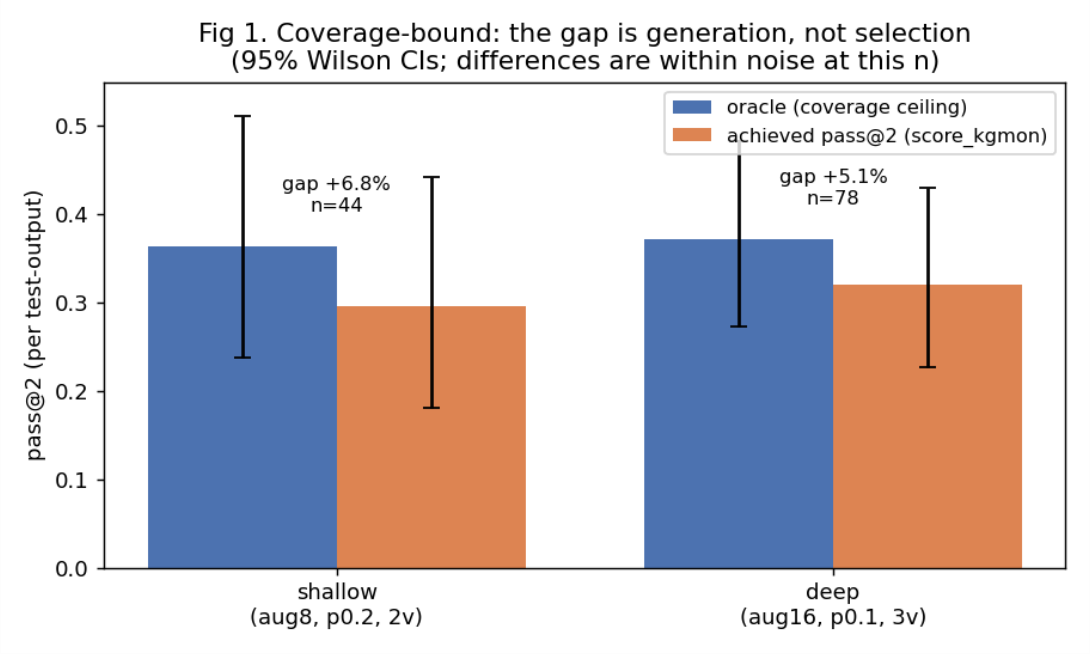


Fig 1. The gap is coverage, not selection: oracle ceiling vs achieved pass@2 for shallow vs deep pools (Wilson 95% CIs; bars are unpaired — different task samples). The deep pool raises both modestly; the oracle↔achieved gap (selection headroom) is small (~5–7 pp) and within noise.

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5. Coverage-bound, not selection-bound

The organizing diagnostic is the gap between **oracle** and achieved pass@2. On the deep pool the oracle is 37.2% and achieved 32.1% (score_kgmon, n=78): a residual *selection* gap of ~5 pp, of which a better fixed selector recovers ≤1 output and a learned verifier at 36× training data recovers one (§4, §9). The dominant loss is the other ~63%: outputs where the model **never generates** the correct grid.

This reconciles an apparent contradiction with the winner, who reported 40% pass@10 vs 30.5% pass@2 (“our scorer missed 10%”). That 10 pp is real **at his pool depth**, but it is a *property of a deeper pool*, not free selection: widening the pool raises pass@N, and only then does selection have something to recover. Our “selection is free money” attempts failed precisely because the answer was absent, not mis-ranked.

A simple model. Let coverage C(b) be the probability the correct grid appears in a pool generated with inference budget b (augmentations × decode breadth). Empirically C grows very slowly in b on ARC-AGI-2 (on the 42 paired outputs, doubling augmentation, halving the decode threshold, and adding views together move the oracle by a net one output — 3 gained, 2 lost; Fig 1 shows the unpaired pools), because the failures are not near-misses the sampler rounds wrong but tasks whose transformation the model cannot represent. Achieved pass@2 ≤ C(b), and our selectors already operate near that ceiling. Hence **score is bounded by C(b), and C(b) saturates** under same-kind compute — exactly the ARC "knowledge/reasoning, not search" thesis, here measured. New coverage requires a *different generation mechanism or a stronger base model*, not more search — consistent with symbolic search (0/167) and offline induction (0/166) adding nothing on the eval distribution.

One exception, and where it stalls: pooling independent runs. Widening *one* run barely moves coverage, but merging the candidate pools of *independent* runs does. On 138 paired public-eval outputs we merged the pools of three local runs (A: the submission config; B: aug_n=16; C: two TTFT epochs) and re-ranked each merged pool with the same score_kgmon selector (`scripts/analyze_pooling.py`):

Pool	Oracle	pass@2	vs A (gained/lost outputs)
A alone	35.5%	31.9%	—
A+B	44.2%	34.1%	+5 / −2 (McNemar p = 0.45)
A+C	44.9%	35.5%	+6 / −1 (p = 0.12)
A+B+C	**48.6%**	36.2%	+7 / −1 (p = 0.07)

Coverage climbs by +13 pp across three runs. TTFT stochasticity reaches transformations that a single run's wider decoding does not, so run-to-run variance is itself a coverage source. But selection converts only about a third of it: the oracle↔pass@2 gap widens from 3.6 pp (one run) to 12.4 pp (three runs). This is the winner's "scorer missed 10%" regime reproduced from our side. Once the pool is deep, **selection becomes the bottleneck**, and our fixed and learned selectors (§4) do not close it. Two costs keep this off the leaderboard. A single pass over the 240 hidden tasks already uses ~10.7 of the 12 h, so a second independent pass means halving each one. And a pool-widening change has inverted on the leaderboard once already (deep

pool: +2.4 pp local → −5.8 pp LB). We report pooling as the clearest remaining headroom, and selection over deep pools as the open problem it exposes. As of Oct 27, 2026 we had not submitted it ourselves; three leaderboard draws of a public cross-run-pooling kernel on the same model (an attributed fork, unmodified apart from our placeholder cell) ran Oct 28–30 and are recorded in `docs/lb_draws.csv`: 30.69, 29.03 and 29.86 (mean 29.86 against the 29.3 same-code median, sd 1.3), inside that series' spread. A public three-pass pooling kernel on the same model therefore scored no better than our single-pass series, consistent with selection rather than coverage being the binding constraint; the kernel's own single-pass baseline was not measured, so this bounds the leaderboard transfer, not the local coverage gain.

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6. The other half: generalization and time

6.1 The eval→LB gap is approach-dependent (Fig 2). From the 2026 forum: the 2025 winner reports LLM+TTT 30.5% eval → 27% public-LB (**−3.5 pp absolute, ~11% relative**); a competitor reports a Tiny Recursive Model 22% → 10.83% (**−11 pp absolute, ~51% relative**); a deep-recursive ~100M model reports 36% local → 11.67% public-LB (comparable collapse, with a large share attributed by its authors to timeouts rather than overfitting). **LLM+TTT is the most LB-robust family on this evidence.** Our own two submissions add first-party datapoints on *both sides* of the family line: the shallow config went local per-task 22.8% → **25.00** (a +2.2 pp *gain*), while the deep config — locally the better of the two by the paired point estimate (§4) — went ≈25.2% → **19.17** (−24% relative). The two points straddle the family calibration (−11% relative), and their spread across two configurations of the *same* system exceeds the between-family differences the calibration rests on: configuration-level choices can dominate family-level robustness. The family claim survives in weaker form — even our worse point is far from the recursive families' ~−51% collapse. A plausible mechanism for the family's robustness (hypothesis, not isolated here): per-task TTT adapts to *this task's* demonstrations, which transfers, whereas small models trained to convergence absorb public-

set statistics. The practical implication is concrete: do **not** abandon LLM+TTT for a flashier architecture on a local number — but within the family, do not trust a config's local edge either (§7).

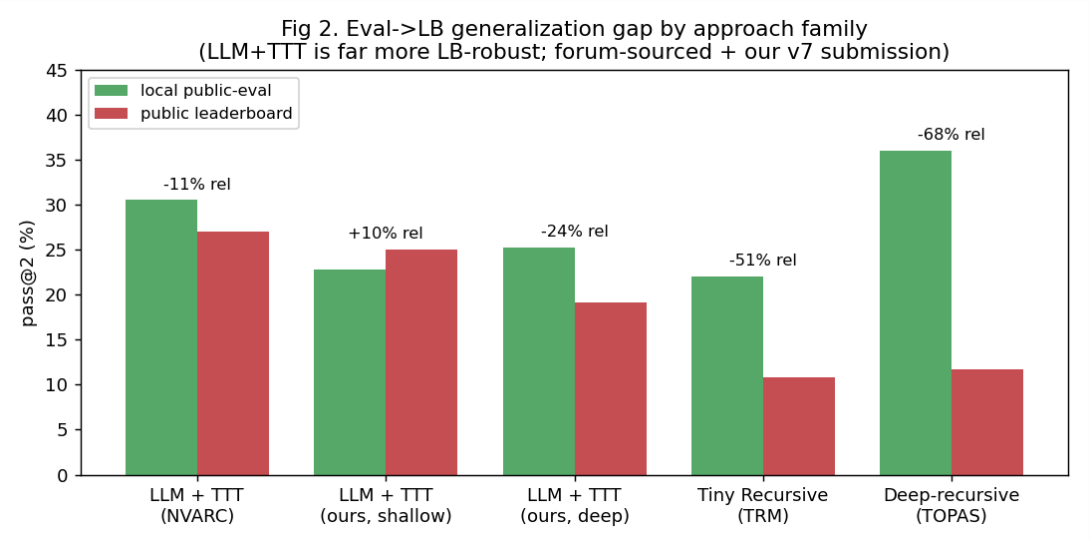


Fig 2. Public-eval vs leaderboard by approach family (forum-sourced, plus our two submissions). Relative drop annotated; our two points straddle the family line.

6.2 Time management is worth large LB points. The 12 h budget is binding. Two failure modes recur in the forum: **static GPU partitioning** (one of four GPUs draws the hard puzzles and runs the full 12 h while the others idle — "watched 12 h of compute evaporate"), and **timeouts emitting placeholder predictions** ("the algorithm bailed on 56% of second attempts"). Our hardened engine (`src/nvarc/kaggle_submit.py`, §8) fixes both with a single **dynamic shared work-queue** (every GPU pulls the next task as it frees, so load self-balances) and a **never-`[[0]]` policy** (any unsolved/unreached output falls back to real guesses — the identity grid and the most-frequent training output, each strictly better than a zero-scoring placeholder). These are engineering, but on the forum evidence they plausibly recover more score than most modeling changes. Our 25.00 shallow submission ran through this engine with **zero placeholder outputs** (4 of 255 outputs fallback-filled) in 3.9 h of the 12 h budget; we cannot isolate the engine's contribution without an ablation submission, but the failure modes it targets did not occur (§9).

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7. What it takes to win, and the cost/quality frontier

Reconciling our measurements with the leaderboard (top ≈ 42.6%) and the 2025 winner's public guidance ("simple tweaks won't be enough — show me 35+"; "this year's environment fits significantly larger models … bigger/newer models will take over"):

- The LLM+TTT recipe **plateaus ~34–35% offline** (the winner's attributed claim); we reach ~33% per output locally and every inference-time lever is saturated (§4–5).
- The path to the winning ~42% is a **stronger base model on harder, compositional synthetic data** — not more search, and, per §4.1, **not parameter scale alone**: an 8B trained on our locally-assemblable corpus lands at half the released 4B's score. The 96 GB environment fits 7–8B models; off-Kaggle training of weights is permitted; the corpus is the part that requires frontier-grade generation.
- **Avoid:** pure symbolic/DSL (we measured 0/167; the winner: "I never believed in DSL for ARC-AGI-2") and pure recursive small models (overfit; large eval→LB gap).

Cost/quality and the 12 h budget (Table 2) — now leaderboard-measured. We submitted both configs. The shallow config scored **25.00** (240 tasks in 3.9 h of the 12 h L4×4 budget; 0 placeholder outputs, 4 fallback-filled). The deep config — locally +2.4 pp paired — scored **19.17**: a **−5.8 pp leaderboard regression**. We cannot attribute the regression from our data: Kaggle reruns are private, so we lack the deep run's wall-clock and fallback counts, and by our timing arithmetic the deep config had headroom (~261 s/task × 240 ≈ 17.4 h single-GPU ≈ 4–5 h wall on the 4-GPU queue). Candidate explanations — none established — are budget pressure we did not instrument, semi-private distribution shift interacting with the wider pool, and run-to-run noise (competitors report ±2 pp on resubmits, short of 5.8 pp). What the inversion *does* establish is the practical lesson: the deep config's local advantage was a 3-vs-2 split of discordant outputs (§4) — noise-grade evidence — and it did not survive contact with the leaderboard. **The**

shallow config is not merely the safe default; on the real leaderboard it dominates. Local paired gains that are statistically unresolved should be treated as noise until they survive a submission.

A recipe-level replication of the inversion. After this paper's first leaderboard results, we found the strongest local effect of the whole study: a TTT-depth change (8 augmentations × 2 epochs instead of 16 × 1, equal token budget) measured **+3.6 pp paired on 138 outputs** (+8/−3 discordant) with a clean dose-response behind it. Deployed runtime-neutrally into the public engine, four independent leaderboard runs drew {27.64, 28.06, 29.31, 27.64} — mean 28.2 vs the unmodified engine's 29.3 over ten draws. **The largest paired local gain we ever measured transferred as ≈ −1 pp.** Together with the deep-pool inversion (+2.4 local → −5.8 LB), the pattern is now replicated: *recipe changes tuned on public-eval pairs overfit the public distribution*; the semi-private set punished both pool-widening and adaptation-deepening. The only interventions that transferred in this study were engine/budget efficiency (+4–5 pp) and placeholder hygiene (≥ 0 by construction).

Time is fully spent, not wasted. An instrumented run of the engine over all 120 public-eval tasks shows its time limits almost never bind: none of 667 decoding calls hit the 540 s cap, 8 of 120 tasks hit the 1,200 s per-task cap (3% of decoding batches skipped), and those 8 tasks solve at the same rate as the rest (25.0% vs 26.5%). Together with a neutral shortest-job-first scheduling test on the leaderboard (61 same-code draws, Aug 25 to Oct 26: mean 29.5, sd 1.3, median 29.3, range 26.4-33.06, vs. 29.3 over ten draws of the unmodified engine; every submission is listed in `docs/lb_draws.csv`), this closes the time lever: the engine's +4-5 pp came from using the whole budget, and there is no remaining budget to reallocate.

Two further submissions complete the picture. First, resubmitting the shallow config *byte-identically* scored 24.17 vs the original 25.00 — one same-code pair differing by 0.8 pp (n=2 runs; competitors report ±2 pp), which the 5.8 pp deep regression comfortably exceeds. Second, an attributed, unmodified fork of the community's public unsloth-based engine (mikelou1's efficiency patch of koushikrudra's baseline, advertised LB 33.89), running the **same sorokin 4B checkpoint** as all our submissions, scored **28.47**: with model, recipe family and scoring held fixed, engine throughput alone — converting the 12 h budget into

per-task test-time compute where our stock-`transformers` port used 3.9 h of 12 — is worth **~+4 pp**. This is the "time" pillar measured with a controlled model. Notably, both engine submissions fell short of the score their source advertises: a fifth submission reran the *unmodified mother baseline* itself (one-line mount-path fix only) and scored **30.14** — our best — against its advertised 33.89. Two independent reruns of the advertised-33.89 family thus land at 28.47 and 30.14: **advertised public-kernel scores overstate rerun expectations by ~4-5 pp** for this budget-sensitive family, a caution that applies equally to the ~34-pp "pack" of leaderboard entries riding these kernels. Our full first-party ladder — 24.17/25.00 (stock `transformers`, one third of the budget used) → 28.47/30.14 (community unsloth engine, full budget, same model checkpoint) — measures the engine/budget lever at **+4 to +5 pp**, the largest single effect in this study.

Table 2 — configs. shallow: aug_n 8, DFS prob 0.2, 2 views, ~130 s/task — **LB 25.00**. deep: aug_n 16, DFS prob 0.1, 3 views, ~261 s/task — **LB 19.17**. Both fit 12 h on L4×4 via the dynamic queue; only shallow fits a single GPU under 12 h.

Table 3 — key hyperparameters. Base Qwen3-4B (16-token grid vocab); per-task LoRA r=256, α=32, dropout 0, target = attn+MLP+embed+lm_head, rslora; TTT lr 5e-5 cosine, 1 epoch over ~128 augmentations, bf16, grad-checkpointing on; decode = batched DFS, keep prob > τ ($\tau \in \{0.2, 0.1\}$); vote = m=8 shared augmentations, score = count − mean(per-aug NLL) or `probmul_3`; max_seq_len 8192. Env: torch 2.11+cu130, transformers 5.9 (the version exhibiting the tokenizer bug), peft 0.19, vllm 0.21 (induction only, with `LD_PRELOAD` of system libstdc++ for Blackwell).

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8. Reproducibility, licensing, attribution

Everything runs on one consumer GPU; hardware: 1× RTX 5090 (32 GB), plus Kaggle L4×4 for submission. Commands:

```
python -m pytest tests/ -
        q                                                    #
        core, no GPU
python scripts/repro_nvarc.py --limit 60 --aug-n 16 --eval-aug-n
        3 --dfs-prob 0.1
```

```
python scripts/analyze_selection.py --pkl outputs/
        cache_deep.pkl             # oracle/selection gap
python scripts/sweep_selection.py  --pkl outputs/
        cache_deep.pkl              # free-selector sweep
python scripts/analyze_pooling.py A=outputs/cache_aug8.pkl
        B=outputs/cache_aug16.pkl C=outputs/cache_ttt_ep2.pkl  #
        cross-run pooling
python scripts/train_verifier.py   --pkl outputs/
        cache_deep.pkl              # learned verifier (group-CV)
python scripts/repro_nvarc.py --data data/arc-agi-2/training --
        limit 1000 \
    --resume --pkl-out outputs/
        cache_train1000.pkl                          # verifier
        training cache
python scripts/
        verifier_transfer.py
        # verifier at scale (train->eval)


python scripts/
        eval_symbolic.py
        # DSL coverage (0/167)


python scripts/induction_complementarity.py --tasks 120 --samples
        48        # BARC net-new (0/166)
python scripts/
        gen_hard_data.py
        # harder-data generator (S4.1)


python -m train.cut_vocab --base Qwen/Qwen3-8B --out models/
        Qwen3-8B_grids16 # 8B vocab cut (S4.1)
PYTHONPATH=src python -m train.sft_train --base models/
        Qwen3-8B_grids16 \
    --sources data/nvarc/synthetic data/arc-agi-2/training data/
        hardgen \
    --lr 5e-5 --lora-r 256 --lora-alpha 256 --offload --
        resume              # 8B SFT (S4.1)
python scripts/
        make_figures.py
        # Fig 1, Fig 2


# offline submission: notebooks/kaggle_submission.ipynb ->
        nvarc.kaggle_submit.main(...)
```

Layout: `src/nvarc/` (ported pipeline + hardened submission engine), `src/arc/` (tokenizer, augmentations, symbolic solver, verifier), `src/induction/` (BARC harness), `src/train/` (SFT: packing, offloaded optimizer, atomic checkpoints; hard-data generator), `src/eval/` (metrics), `scripts/` (every experiment), `data/` (gitignored; fetched at runtime).

Submitted kernel. The linked 57502566 and the pinned daily series of §7 ran version 4 of the Kaggle notebook `arc-agi2-baseline-fallback`; its exact source and metadata are in `kaggle_kernels/baseline-fallback/`. Relative to the attributed public engine it contains mount-path fixes, the shortest-job-first task queue in `starter.py` (tasks sorted by grid-cell count instead of alphabetically), our never-[[0]] fallback cell (§6.2), and the scaffolding of a verified-induction pre-pass ("Leg C") that exits as a no-op in this configuration because no coder model is attached (the notebook attaches only the 4B checkpoint), so the merge step is skipped and the base pipeline runs as in the public kernel apart from the queue order. The cross-run-pooling fork that ran Oct 28–30 is in `kaggle_kernels/triple-pass-fork/`.

Licensing & attribution. Original files are MIT-0 (see `LICENSE`). Vendored/derived files retain upstream licenses and carry attribution headers: `src/nvarc/arc_loader.py` and `src/nvarc/selection.py` derive from NVARC (1ytic/NVARC, open-sourced per the ARC Prize requirement); `src/induction/common.py` is vendored from BARC (xu3kev/BARC). We **do not redistribute** model weights, synthetic datasets, or the competition forum corpus: all of `data/` is gitignored and fetched at runtime from Kaggle/HuggingFace. The forum corpus is used only for the quantitative analysis of §6 and is cited, not relicensed.

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9. Limitations and threats to validity

- **Four submissions, a thin distribution.** Our identical-code pair (25.00 / 24.17) gives one first-party estimate of same-code rerun noise (0.8 pp, n=2) alongside competitors' reported ±2 pp; the *sign* of the 5.8 pp deep-config regression is robust to either figure, its magnitude less so. The engine-efficiency estimate (+4 to +5 pp) rests on two runs of ours (24.17, 25.00) against the community engine's 71 leaderboard draws (median 29.3, mean 29.5; §7): the engine side is well sampled, our side is not. Grand-Prize eligibility (a linked Kaggle submission) is satisfied.
- **Small samples.** Most A/Bs are n=20–78; as Table 1 states, sub-~10 pp effects are within CI. The robust claims are the *true zeros* (DSL 0/167, induction 0/166 eval) and the *pattern* of uniformly-

unresolved positive effects; the +2.4 pp deep-pool and the verifier's ±1.3 pp are point estimates within noise, not established effects.

- **Verifier data scale — now tested.** Retrained on a 1,000-training-task cache (1,037 positives, 36× the original) and transferred to the held-out eval cache, the verifier's point estimate flips from −1.3 pp to +**1.3 pp** (26 vs 25 of 78) — one task, within noise, and its dominant feature is the augmentation-NLL rank the fixed scorer already uses (`scripts/verifier_transfer.py`). Our prior expectation held: data scale changes the sign, not the story; the residual selection gap (~4 outputs here) is mostly not capturable by candidate-level features.
- **The training-time results are single-GPU-scale.** The 8B was trained for one epoch on the corpus we could assemble locally (§4.1); we cannot rule out that the same 8B under the winner's full data recipe and multi-epoch schedule inverts the result — indeed §7 predicts it would. What our null establishes is narrower and useful: at *equal, locally-feasible* data, parameter scale subtracts rather than adds. The winning path (§7) still requires cloud compute outside this study.
- **Forum-sourced numbers** (§6) are self-reported by competitors; we attribute each to its source and treat the mechanism explanations as hypotheses.

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10. Related work

Test-time training for ARC was established by Akyürek et al. ("The Surprising Effectiveness of Test-Time Training for Abstract Reasoning"); NVARC (2025) combined it with heavy synthetic-data generation and augmentation voting to win the offline track. Program-induction approaches (BARC; "Combining Induction and Transduction") generate and execute Python solvers; Tiny Recursive Models (Jolicoeur-Martineau, 2025) pursue tiny recurrent reasoners. Our delta is not a new solver but a *measured decomposition* of where the offline score is won and lost — coverage vs selection, and generalization vs local accuracy — across these families, on the harder ARC-AGI-2 distribution.

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11. Conclusion

The offline ARC-AGI-2 score is governed by coverage, generalization, and time more than by decoding/selection cleverness. We reproduced the winning recipe on a single GPU, showed via N+CI measurements that neither inference-time levers (~33% per-output locally, saturated) nor locally-feasible training-time levers (harder local data: ±0.0 pp; an 8B on the same corpus: directionally *worse*) resolvably move the score, calibrated the eval→LB gap that makes LLM+TTT uniquely robust — then confirmed it on the leaderboard, where our shallow config scored 25.00 with a slightly positive gap and our locally-positive deep config regressed by 5.8 pp. The path past ~35% is a stronger base on better synthetic data — where "better" is beyond what a single GPU can generate or absorb — not more search and not more parameters. Our most useful contribution is the measured map of the dead ends, now with two of its edges leaderboard-verified.

References (selected)

- ARC Prize 2025 results & technical report; ARC-AGI-2 technical report (arcprize.org).
- NVARC 2025 winning solution and code (github.com/1ytic/NVARC).
- Akyürek et al., *The Surprising Effectiveness of Test-Time Training for Abstract Reasoning*.
- Jolicoeur-Martineau, *Tiny Recursive Models* (2025).
- *Combining Induction and Transduction for Abstract Reasoning* (BARC; github.com/xu3kev/BARC).
- ARC Prize 2026 Kaggle competition discussion forum (analyzed in §6; not redistributed).